



INFORMATICS INSTITUTE OF TECHNOLOGY
In Collaboration with
UNIVERSITY OF WESTMINSTER

EXPLAINABLE AI-POWERED LEGAL CONTRACT INTELLIGENCE SYSTEM FOR RISK ASSESSMENT AND TEMPORAL COMPLIANCE MONITORING

A Final Year Project Proposal

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Abstract

Commercial contracts increasingly govern activities that are subject to fast changing statutory and regulatory obligations, yet contract review in practice remains a largely manual process that struggles to keep pace with legislative change. Existing artificial intelligence tools for contract analysis can classify clauses and highlight risk, but they typically perform a single, point in time assessment and rarely explain a flagged risk with reference to the specific, dated legal provision that justifies it. As a result, a clause that is compliant on the day of signing can silently become non compliant once the underlying law or regulation is amended, without any mechanism to alert the contracting parties.

This proposal investigates an explainable AI powered legal contract intelligence system that performs clause level risk assessment and continuously monitors contracts for temporal compliance drift against a versioned regulatory knowledge base. The system combines transformer based clause classification, trained and evaluated on established annotated contract corpora, with an explanation layer that links each risk flag to the exact statutory or regulatory provision, including its effective date, and with a temporal monitoring module that re evaluates existing contracts whenever a linked law or regulation is amended. The regulatory knowledge base draws on publicly available Sri Lankan legislative and regulatory document collections, including Acts, Bills and Extraordinary Gazettes, while contract level training and evaluation rely on internationally recognised annotated contract datasets. Evaluation will combine clause level classification performance, explanation faithfulness and stability testing, and dedicated experiments that measure how reliably the system detects compliance drift after a simulated legal amendment. The expected contribution is an evaluated framework and open experimental evidence showing whether linking legal explanations to versioned, dated regulatory provisions improves the reliability and legal usefulness of automated contract risk assessment.

Keywords: explainable AI; legal contract analysis; contract risk assessment; temporal compliance monitoring; legal natural language processing; regulatory knowledge base; clause classification.

Declaration and Generative AI Disclosure

I declare that this proposal is my own work, that it has not been submitted previously for any other academic award, and that all sources consulted have been properly acknowledged through in-text citation and full reference in accordance with the Harvard referencing convention.

Generative AI disclosure. Generative AI tools were used in a limited, supporting capacity while preparing this proposal. They were used to brainstorm alternative literature search terms, to check the clarity of sentences that I had already drafted, and to suggest structural checklists that I then adapted to my own project. They were not used to invent citations, fabricate results, or generate the research design. Every cited source listed in the References section corresponds to a document that I have personally read, either as a PDF supplied in my project research folder or as an openly accessible dataset or paper page, and I have verified the title, authorship and year before citing it. No confidential data, restricted institutional material, or another person's unpublished work was submitted to any Generative AI tool. I remain fully responsible for the content, structure, argument and conclusions presented in this document.

Signature:  Date: 01/09/2026

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List of Abbreviations

Abbreviation	Meaning
AI	Artificial Intelligence
XAI	Explainable Artificial Intelligence
NLP	Natural Language Processing
CUAD	Contract Understanding Atticus Dataset
LLM	Large Language Model
LEDGAR	Labeled EDGAR Contract Provisions Corpus
NLI	Natural Language Inference
GDPR	General Data Protection Regulation (EU) 2016/679
AIA	Artificial Intelligence Act, Regulation (EU) 2024/1689
PDPA	Personal Data Protection Act, No. 9 of 2022 (Sri Lanka)
ETA	Electronic Transactions Act, No. 19 of 2006 (Sri Lanka)
LESP	Legal, Ethical, Social and Professional (issues)
RQ	Research Question
RO	Research Objective
FR/NFR	Functional / Non-Functional Requirement
API	Application Programming Interface
ECE	Expected Calibration Error
SHAP	SHapley Additive exPlanations
LIME	Local Interpretable Model-agnostic Explanations
CI	Continuous Integration

Chapter 1 - Introduction

1.1 Chapter overview

This chapter introduces the problem of keeping automated legal contract analysis aligned with a constantly evolving body of statutes and regulations. It begins by placing the project within the broader domain of legal artificial intelligence, then narrows the discussion to the specific technologies involved, namely clause level classification, explainability and regulatory change tracking, and then connects the technical problem to the practical

workflow of legal and compliance professionals. The chapter proceeds to define the precise problem being addressed, states the motivation for the work, analyses the gaps left open by existing research and tools, and converts those gaps into contributions, research questions, an aim and a set of measurable objectives. It closes with the scope of the project, the expected deliverables and a short summary that leads into the literature review in Chapter 2.

1.2 Problem background

1.2.1 Research domain context

Legal artificial intelligence has matured considerably over the past decade, moving from simple keyword based document search toward systems that can classify, summarise and reason over legal text. Within this domain, contract analysis is one of the most commercially active applications, because contracts are high volume, high stakes documents whose review has traditionally depended on the time of qualified lawyers. Practitioner facing tools already offer automated clause extraction and risk scoring (Gangadhar, 2026; Prakash, Ramesh and Nair, n.d.), and academic work continues to examine how AI systems interact with the legal obligations that surround them, including contract law and tort law (Hacker et al., 2020). At the same time, explainability has become a central requirement rather than an optional feature, because decisions that affect legal rights must be justifiable to the people they affect (Saeed and Omlin, 2021; Longo et al., 2024). This project sits at the intersection of these two developments, treating contract intelligence and explainability as a single, connected problem rather than two separate features.

1.2.2 Specific technological context

The proposed work draws on three connected technical strands. The first is clause level representation and classification, typically implemented with transformer encoders that have been adapted to legal language, such as Legal-BERT style models, and evaluated on annotated contract corpora including the Contract Understanding Atticus Dataset (Hendrycks et al., 2021) and the LEDGAR provision corpus (Tuggener et al., 2020). The second is explainability, which in the legal setting must go beyond a visually plausible heatmap and instead produce explanations that a professional could use to contest or defend a decision, a requirement that recent human-computer interaction and argumentation research has made explicit (Mansi, Karusala and Riedl, 2025; Prajescu and Confalonieri, 2025). The third strand, and the one that is least developed in the reviewed literature, is temporal regulatory awareness, meaning the ability of a system to recognise

that a legal provision it relied on for an earlier assessment has since been amended. Sri Lankan public legal document collections, including Acts, Bills and Extraordinary Gazettes (Senaratna et al., 2026), provide a realistic, versioned regulatory source that can be used to build and test this temporal capability.

1.2.3 Practical and real-world context

In everyday legal practice, an in-house counsel or compliance officer reviewing a supplier or service contract must check each clause against the law as it stands at the time of review, and must periodically revisit signed contracts whenever the underlying law changes, for example when a new gazette introduces a stricter data protection or consumer protection requirement. This second step, ongoing monitoring of already signed contracts, is rarely done systematically because it requires continuous legal awareness that is expensive to maintain manually. A decision support tool that only classifies risk at the point of signing addresses half of this workflow. The system proposed here is deliberately framed as a decision support aid that keeps a human reviewer in the loop, flags clauses for attention together with the specific legal basis for the flag, and separately signals when a previously reviewed clause may need re-examination because a linked regulation has changed, rather than issuing an automatic verdict on legal validity.

1.3 Problem definition

Existing AI powered contract analysis tools can identify risky clauses and, in some cases, check them against a fixed snapshot of applicable regulations (Prakash, Ramesh and Nair, n.d.; Gangadhar, 2026), but they generally treat compliance as a static property of the clause rather than a relationship between the clause and a specific, dated version of a law. Explainability research in the legal domain has established that legal explanations need to be actionable and contestable (Mansi, Karusala and Riedl, 2025) and that argumentation based approaches can align explanations with regulatory reasoning (Prajescu and Confalonieri, 2025), yet this reasoning has not been connected to a mechanism that tracks when the underlying regulation itself changes. The problem addressed by this research is therefore twofold: producing clause level risk explanations that are traceable to a specific legal provision, and maintaining the validity of those explanations over time as the referenced provisions are amended, repealed or replaced. The project is scoped to English-language commercial contract clauses and to regulatory sources that are publicly available in a structured, dateable form; it does not attempt to provide binding

legal advice, to cover every area of law, or to replace qualified legal review.

1.3.1 Problem statement

Current AI powered contract analysis tools do not adequately combine clause level risk explanation that is traceable to a specific, dated legal provision with a mechanism for detecting when that provision changes, leaving reviewed contracts vulnerable to undetected temporal compliance drift.

1.4 Research motivation

The technical motivation for this project is the observation that clause classification accuracy and explanation quality are usually evaluated separately from regulatory currency, even though all three properties are needed before an explanation can be trusted by a legal professional. The practical motivation is that compliance drift, where a contract silently becomes non compliant after a law changes, is a genuine and costly risk for organisations operating in fast changing regulatory environments, and Sri Lanka's frequent use of Extraordinary Gazettes to introduce urgent regulatory changes makes this risk concrete and testable using real, openly available data (Senaratna et al., 2026). The academic motivation is that connecting contract level explainable AI with temporal regulatory tracking has not been evaluated together in the reviewed literature, creating an opportunity to produce new comparative evidence rather than simply combining existing components. Finally, working at the boundary of computing and law is professionally relevant to a computing graduate who intends to work in legal technology, and the supervisor's dual background in computing and law provides an unusual opportunity to ground the technical design in genuine legal reasoning rather than superficial legal terminology.

1.5 Research gap analysis

The reviewed literature and practitioner tools point to four specific, addressable gaps. Gap one concerns the disconnect between legal explainability research and contract risk tools: work on legally informed and argumentation based explanation establishes what a good legal explanation should contain (Mansi, Karusala and Riedl, 2025; Prajescu and Confalonieri, 2025), but existing contract risk analysers do not appear to implement explanations at this level of legal traceability (Gangadhar, 2026; Prakash, Ramesh and Nair, n.d.). Gap two concerns temporal awareness: reviewed regulatory compliance checkers and contract analysers perform what is effectively a snapshot assessment and do not model the fact that the applicable law can change after the contract has been reviewed. Gap three concerns

evaluation of legal explanations, since XAI surveys note that faithfulness and stability testing remain uneven across application domains (Saeed and Omlin, 2021; Kostopoulos et al., 2024; Longo et al., 2024), and this unevenness is likely to be more severe for a narrow, high-stakes task such as clause level legal risk explanation. Gap four concerns data integration: Sri Lankan legislative and regulatory corpora exist at a large scale (Senaratna et al., 2026) but are not linked, at clause level, to contract risk categories, which means that a working temporal compliance system must build this linkage rather than rely on an existing labelled resource.

Gap	Evidence from literature and practice	Planned response	Linked RQ	Linked RO
G1	Legal explainability principles are defined in HCI and argumentation research but not implemented at clause level in contract risk tools	Explanation module that cites the specific dated regulatory provision behind each risk flag	RQ2	RO3, RO4
G2	Contract analysers and compliance checkers perform point-in-time, snapshot assessment only	Temporal compliance monitor that re-checks clauses against a versioned regulatory knowledge base	RQ3	RO4, RO5
G3	Faithfulness and stability evaluation of legal explanations is inconsistent across the wider XAI literature	Dedicated faithfulness, stability and legal-traceability evaluation protocol	RQ2	RO5

G4	Sri Lankan regulatory corpora are large but not linked to contract clause risk categories	Purpose-built linkage layer connecting clauses to dated Acts and Gazette provisions	RQ3, RQ4	RO2, RO3

Table 1.1: Research gap analysis matrix

1.6 Contribution to the body of knowledge

1.6.1 Research-domain contribution

The research domain contribution will be an integrated experimental framework that evaluates clause level risk classification, legally traceable explanation generation and temporal compliance monitoring as a single system rather than as separate modules. The project will also produce comparative evidence on which explanation methods remain faithful and stable when applied specifically to legal risk predictions, and an ablation study showing how much of the system's practical value comes from the temporal monitoring component rather than from clause classification alone.

1.6.2 Problem-domain contribution

The problem domain contribution will be a working prototype that a compliance reviewer could use to obtain a clause level risk flag together with the exact statutory or regulatory clause that justifies it, and a separate, clearly distinguished alert when a previously reviewed contract is affected by a subsequent legal amendment. This directly targets the practical gap between one-off contract review and ongoing compliance monitoring described in Section 1.2.3.

1.7 Research challenges

Several challenges must be managed for this project to succeed. The first is the absence of a publicly available, annotated Sri Lankan contract dataset, which means that clause classification must be trained primarily on international English-language contract corpora and only validated, at a small scale, against local example contracts. The second is linking contract clauses to specific statutory or regulatory provisions in a way that is accurate enough to be legally meaningful, which will require a combination of retrieval techniques and manual verification of the most important links. The third is simulating regulatory change realistically, since the project timeline will not span an actual multi-year legal

amendment cycle, so amendments will need to be reconstructed from the dated version history already present in the Acts and Extraordinary Gazette collections. The fourth is keeping the explanation faithful to the model rather than merely convincing, which will require perturbation based faithfulness testing rather than relying on the visual plausibility of highlighted text. The fifth is computational feasibility within the available project timeline and hardware, which will be managed through staged experiments and efficient baseline models before any larger model is attempted.

1.8 Research questions

RQ1 How accurately can transformer-based clause classification models identify contract risk categories compared with simpler baseline approaches, when trained and evaluated on established annotated contract corpora?

RQ2 Which explanation method produces the most faithful and stable clause-level legal risk explanations, and how well can each explanation be traced to a specific, dated regulatory provision?

RQ3 How reliably can a temporal compliance monitoring module detect that a previously assessed contract clause is affected by a subsequent amendment to a linked statute or regulation drawn from the Sri Lankan legal document collections?

RQ4 Does presenting a risk explanation together with the specific dated regulatory provision improves its perceived legal usefulness compared with a generic, provision-free explanation, based on structured feedback from an available legal reviewer where access permits?

1.9 Research aim

The aim of this research is to design, develop and evaluate an explainable AI powered legal contract intelligence system that performs clause level risk assessment and continuously monitors contracts for temporal regulatory compliance drift, producing explanations that are traceable to specific, dated statutory and regulatory provisions.

1.10 Research objectives

RO1 Critically analyse the limitations of existing contract risk tools, legal explainability

research and regulatory tracking approaches relevant to temporal compliance monitoring.

RO2 Define the functional, non-functional and data requirements for a clause-level, explanation-aware, temporally-aware contract intelligence system.

RO3 Design a modular architecture separating clause classification, explanation generation and temporal compliance monitoring so that each component can be evaluated and replaced independently.

RO4 Implement the proposed system together with suitable baselines, including a non-explainable classifier and a snapshot-only compliance checker.

RO5 Evaluate predictive performance, explanation faithfulness and stability, and compliance-drift detection accuracy, using ablation studies to isolate the contribution of each module.

RO6 Analyse the legal, ethical and professional implications of the proposed system and define the conditions under which it can be responsibly used as a decision support aid.

1.11 Scope, delimitations and assumptions

This project will focus on English-language commercial contracts and will use the Contract Understanding Atticus Dataset, the LEDGAR provision corpus and ContractNLI as the primary sources of annotated clause-level training and evaluation data (Hendrycks et al., 2021; Tuggener et al., 2020; Koreeda and Manning, 2021). The regulatory knowledge base will be built from the English-language subset of the Sri Lankan Acts, Bills and Extraordinary Gazette collections (Senaratna et al., 2026), used to reconstruct realistic, dated legal amendments for temporal compliance testing. A small, separately labelled set of local example contracts will be used only for illustrative, small-scale validation and will not be presented as a large annotated Sri Lankan contract corpus. The project will not provide binding legal advice, will not attempt full multilingual coverage of Sinhala and Tamil legal text in this phase, and will not claim readiness for production legal use. It assumes that the public dataset repositories remain accessible for the duration of the project and that a representative subset of regulatory documents is sufficient to demonstrate the temporal monitoring mechanism without processing the full multi-gigabyte collections.

1.12 Expected outcomes and deliverables

- A cleaned and linked data pipeline connecting contract clause categories to specific Sri Lankan Acts and Extraordinary Gazette provisions.
- Baseline and proposed clause classification models with documented training configurations.
- An explanation module that returns a clause-level risk flag together with its supporting regulatory citation.
- A temporal compliance monitoring module capable of re-evaluating previously reviewed contracts after a simulated regulatory amendment.
- An evaluation report covering classification performance, explanation faithfulness and stability, and compliance-drift detection accuracy.
- A working demonstrator interface, an experiment configuration repository, and the final dissertation documenting the complete research process.

1.13 Chapter summary

This chapter established that reliable AI powered contract review requires more than accurate clause classification: it requires explanations that are traceable to specific legal provisions and a mechanism for recognising when those provisions change. Four gaps were identified across legal explainability research, practitioner contract tools and Sri Lankan regulatory data, and these were converted into a research aim, four research questions and six objectives centred on clause classification, legally traceable explanation and temporal compliance monitoring. Chapter 2 now critically reviews the relevant literature, datasets and evaluation practices needed to justify the proposed architecture and methodology described in Chapter 3.

Table 1.2: Chapter 1 alignment checkpoint

Element	Control question	Primary location
Problem	What precise limitation is being solved?	Section 1.3
Gap	What does existing work not address adequately?	Section 1.5
Contribution	What new evidence or artefact responds to the gap?	Section 1.6
Question	What must the study answer?	Section 1.8
Objective	What activity will answer the question?	Section 1.10

Method	How will the activity be performed?	Chapter 3
Evidence	What metric or analysis will demonstrate the answer?	Section 3.9

Chapter 2 - Literature Review

2.1 Chapter overview

This chapter reviews the domain requirements of legal contract analysis before examining the technology pipeline used to build clause level risk models, the explanation methods available for legal decision support, and the datasets and evaluation practices that will underpin the proposed system. The review is organised thematically, comparing groups of related studies rather than summarising one paper at a time. It closes with a synthesis that positions the proposed research against the closest existing work and presents the conceptual architecture that the methodology in Chapter 3 will implement.

2.2 Review strategy and concept map

2.2.1 Literature search and selection strategy

The literature base for this proposal combines a targeted academic search with a curated project research folder assembled during the early scoping stage of the project. Search terms combined variants of explainable AI, legal AI, contract analysis, regulatory compliance and temporal or dynamic compliance, applied across arXiv, general web search and the supervisor recommended reading list. Studies were included where they presented a method, dataset, evaluation protocol, or legal analysis directly relevant to contract level explainability or regulatory compliance, and where sufficient methodological or legal detail was available to assess relevance. Editorials without technical or legal substance were excluded. All items cited in this proposal were retained and read as PDF or web sources within the project folder, and their bibliographic details were verified against the source file or its official page before citation.

2.2.2 Concept map

Figure 2.1 places reliable, explainable and temporally aware contract review at the centre of the review. Three branches radiate from this centre: contract clause modelling, legal explainability, and regulatory and compliance tracking, with data sourcing and evaluation shown as supporting layers that connect all three branches to the proposed contribution.

2.3 Problem-domain review

2.3.1 Domain task and decision context

The target task is clause level risk classification supported by an explanation and a temporal compliance signal, rather than an end to end legal judgement. Inputs available in practice

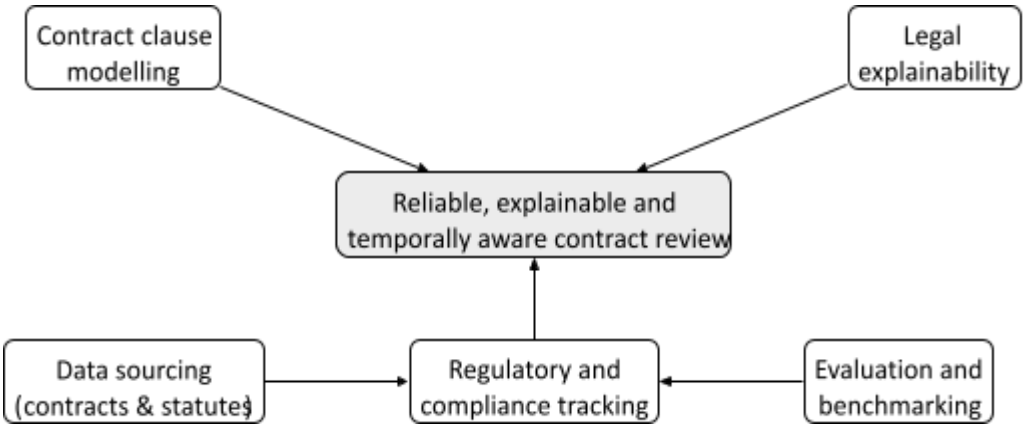


Figure 2.1: Concept map linking contract modelling, legal explainability and temporal compliance tracking

include the contract text itself, the identity of the governing jurisdiction, and, where accessible, the applicable statutory and regulatory framework in force at review time. Errors in this domain are asymmetric: a missed risk flag can expose an organisation to a legal or financial liability that may not surface for months or years, whereas a false alarm mainly costs reviewer time. This asymmetry, discussed in general terms by Hacker et al. (2020) in relation to contract and tort law liability, means that the evaluation strategy in this project cannot rely on overall accuracy alone and must examine class specific error behaviour together with the human reviewer's ability to interpret and act on each flag.

2.3.2 Data and operational challenges

Contract review in practice is complicated by inconsistent clause wording across drafting styles, the presence of boilerplate language that resembles risky clauses without carrying the same legal weight, and the fact that the same clause category can be treated differently under different jurisdictions. Regulatory data adds a further layer of difficulty because Sri Lankan Acts, Bills and Extraordinary Gazettes are published as long, unstructured legal documents rather than as pre-segmented, machine readable provisions (Senaratna et al., 2026), and the same subject can be addressed piecemeal across an Act and several later

Gazette amendments. These properties motivate careful clause and provision segmentation during preprocessing and a conservative approach to linking clauses to regulatory text.

2.3.3 Standards, regulations and professional expectations

Any contract intelligence system that processes organisational or personal data operates under Sri Lanka's Personal Data Protection Act, No. 9 of 2022, which governs the collection and processing of personal data, and under the Electronic Transactions Act, No. 19 of 2006, which establishes the legal recognition of electronic contracts and records. At a broader level, the European Union's General Data Protection Regulation and the Artificial Intelligence Act (Regulation (EU) 2024/1689) illustrate the kind of transparency and accountability expectations that increasingly apply to automated decision support tools, even where the project's primary jurisdiction is Sri Lanka rather than the European Union. Professional expectations from the legal explainability literature reinforce this position: explanations offered to affected stakeholders should be actionable and contestable rather than merely descriptive (Mansi, Karusala and Riedl, 2025).

2.4 Technological review by implementation stage

2.4.1 Data acquisition and curation

Clause level training data will be drawn from the Contract Understanding Atticus Dataset, comprising 510 commercial contracts with more than 13,000 expert annotations across 41 clause categories under a CC BY 4.0 licence (Hendrycks et al., 2021), and from LEDGAR, a large multi-label corpus of contract provisions extracted from public United States Securities and Exchange Commission filings (Tuggener et al., 2020). ContractNLI provides a complementary resource for verifying whether a clause entails, contradicts, or is silent on a compliance-relevant hypothesis (Koreeda and Manning, 2021). Regulatory data will be drawn from the Sri Lanka Acts, Bills and Extraordinary Gazette collections, all released under an MIT licence and accessible through GitHub and Hugging Face (Senaratna et al., 2026). Public, English-language, permissively licensed sources were preferred over restricted or unverified alternatives to keep the data pipeline reproducible.

2.4.2 Pre-processing and quality control

Contract documents will be segmented into individual clauses using sentence and paragraph boundary heuristics validated against the existing CUAD and LEDGAR annotations, and normalised to remove formatting artefacts such as page headers and signature blocks. Regulatory documents will be segmented into sections and subsections, and each will be

tagged with an effective date derived from the enactment or gazette publication date. Because both corpora contain repeated or near duplicate text, exact and near duplicate detection will be applied before splitting data into training, validation and test partitions to prevent leakage.

2.4.3 Representation and feature extraction

Clause representation will use a pretrained transformer encoder fine-tuned on legal text, following the general direction supported by recent decision-support and explainability reviews that favour transformer-based text encoders for structured legal classification tasks (Kostopoulos et al., 2024). A smaller TF-IDF or classical machine learning baseline will also be implemented, consistent with Hacker et al. (2020), who compared several classical models against the explainability-accuracy trade-off, to provide an interpretable reference point against which the transformer model's added value can be measured.

2.4.4 Regulatory and temporal provision modelling

Each regulatory provision extracted during preprocessing will be represented with its text, its effective date, and, where an amendment exists, a link to the provision it replaces. This creates a simple version chain per provision. Retrieval between a contract clause and its most relevant regulatory provision will use dense embedding similarity combined with keyword filtering on defined legal subject tags, allowing the system to identify not only the currently applicable provision but also any earlier version that governed the clause at the time of the original contract review.

2.4.5 Risk-confidence calibration and escalation

Because a legal risk flag should not be presented with unwarranted certainty, the classification head will be calibrated using temperature scaling, and a confidence threshold will determine whether a clause is flagged automatically, flagged for mandatory human review, or passed without comment. This mirrors the general calibration and selective-prediction practice recommended in reliability-focused decision-support reviews (Kostopoulos et al., 2024), adapted here to a legal risk-escalation rather than a purely diagnostic setting.

2.4.6 Explainability methods

Two complementary explanation strategies will be compared. The first is a post-hoc attribution method, such as SHAP or attention-based highlighting, that identifies which words or phrases in the clause most influenced the risk prediction. The second is an

argumentation-informed explanation that expresses the flag as a structured claim linking the clause to the specific regulatory provision that supports it, following the direction proposed by Prajescu and Confalonieri (2025) for legally relevant, contestable explanations, and reflecting the actionable and contestable explanation criteria set out by Mansi, Karusala and Riedl (2025). The project will evaluate whether the argumentation-informed explanation is judged more legally useful than the attribution-only explanation.

2.4.7 Hyperparameter optimisation and training strategy

Model selection will use a bounded random search over learning rate, batch size and classification threshold, evaluated only on the validation partition, with early stopping based on validation macro-F1. Class imbalance across the 41 CUAD categories will be addressed using class-weighted loss and, where necessary, oversampling of minority risk categories. All experiment configurations, random seeds and results will be logged to support reproducibility.

2.5 Existing work: thematic analysis

Theme 1, explainability and legal liability. Hacker et al. (2020) show, through legal case studies and a machine learning experiment on an SMS spam dataset, that explainability is relevant not only to data protection law but also to contract and tort law liability, and that professionals may need to balance model accuracy against interpretability to meet a legal standard of care. Their analysis provides the legal justification for treating explainability as a first-class requirement in this project, although their study does not implement a contract analysis system, use a legal contract dataset, or address temporal compliance, which is precisely the space this project occupies.

Theme 2, general explainable AI challenges. Saeed and Omlin (2021) and Longo et al. (2024) both catalogue open challenges in explainable AI across the design, development and deployment stages of the machine learning lifecycle, including the need for domain-specific faithfulness evaluation and better alignment between explanations and stakeholder needs. Kostopoulos et al. (2024) review explainable decision support systems more broadly and highlight recurring weaknesses in how explanation quality is measured. None of these surveys focus on legal contract review specifically, but together they establish that faithfulness and stability testing, rather than visual plausibility alone, must be central to the evaluation plan in this project.

Theme 3, legally informed and argumentation-based explanation. Mansi, Karusala and Riedl (2025) argue for explanations that are actionable and contestable and identify distinct stakeholder groups with different explanation needs in high-stakes legal contexts. Prajescu and Confalonieri (2025) extend this direction by proposing argumentation-based explanation

as a way of capturing the defeasible, contestable nature of legal reasoning and aligning explanations with regulatory frameworks such as the GDPR and the EU AI Act. Mafleh et al. (2026) examine the broader legal impact of AI on contract interpretation and execution, underlining those courts and regulators are actively grappling with how AI-assisted decisions should be treated in contract disputes. Together these studies support the choice of an argumentation-informed explanation format in this project, but none of them connects such an explanation to a live, versioned regulatory knowledge base.

Theme 4, applied contract and compliance tools. Practitioner and applied research describes systems that already perform contract risk analysis or regulatory compliance checking, including an AI-powered legal contract risk analyser (Gangadhar, 2026), an AI-powered regulatory compliance checker for contracts (Prakash, Ramesh and Nair, n.d.), and broader discussions of generative AI and NLP applied to legal risk management, compliance and contract analysis decision-making (Anon., n.d.). A further practitioner discussion piece considers evidence-based AI risk assessment in the public policy setting, illustrating that structured, evidence-linked risk assessment is a live concern beyond the contract domain as well (Anon., 2026). These applied systems demonstrate technical feasibility for clause-level risk scoring, but based on the available material they operate as point-in-time checkers and do not appear to re-assess previously reviewed contracts when the underlying law changes, which is the specific gap this project addresses.

2.6 Dataset review and comparison

Table 2.1 compares the datasets that will support this project. Contract-level data will come from established, expert-annotated international corpora, since no comparably annotated Sri Lankan contract corpus currently exists in the public domain. Regulatory data will come from Sri Lankan legislative and regulatory collections, which are large, dated and permissively licensed, making them well suited to reconstructing realistic temporal compliance scenarios even though they were not originally built for this purpose.

Table 2.1: Dataset comparison for contract modelling and regulatory tracking

Dataset	Scale	Licence	Content	Planned role
CUAD (Hendrycks et al., 2021)	510 contracts, 13k+ annotations	CC BY 4.0	41 expert-labelled clause categories	Primary clause classification training/evaluation
LEDGAR (Tuggener et al., 2020)	~846k provisions	CC BY 4.0	Multi-label SEC contract provisions	Large-scale supplementary classification data

ContractNLI (Koreeda and Manning, 2021)	607 NDAs	CC BY-NC	Entailment labels vs. 17 hypotheses	Compliancestyle clause verification
Sri Lanka Acts (Senaratna et al., 2026)	3,976 docs, 6.8 GB	MIT	Enacted legislation, 1981–2026	Primary regulatory knowledge base
Sri Lanka Extraordinary Gazettes (Senaratna et al., 2026)	~105k docs	MIT	Regulations and urgent notices, 2010–2026	Source of dated regulatory amendments
Sri Lanka Bills (Senaratna et al., 2026)	4,209 docs	MIT	Draft legislation, 2010–2026	Forwardlooking regulatory change context
Sri Lanka Court Judgments (Senaratna et al., 2026)	~19k docs	MIT	Appeal and Supreme Court decisions	Supporting interpretive evidence (optional)

2.7 Evaluation and benchmarking review

Benchmarking practice in the reviewed explainable AI literature stresses that predictive performance alone is an incomplete measure of a trustworthy decision-support system, and that calibration, selective prediction and explanation quality all need dedicated metrics (Saeed and Omlin, 2021; Kostopoulos et al., 2024; Longo et al., 2024). For clause classification, this project will report macro-F1 and class specific precision and recall across risk categories rather than a single aggregate accuracy figure, because the clause categories in CUAD and LEDGAR are heavily imbalanced. For explanations, faithfulness will be tested with a deletion or insertion perturbation protocol, and stability will be tested by measuring how much an explanation changes under small, meaning-preserving edits to the clause text. For the temporal compliance component, a dedicated protocol will simulate a regulatory amendment by substituting an earlier, dated version of a linked provision and measuring whether the system correctly re-flags the affected clauses, reported as a drift-detection precision and recall pair rather than a single score.

2.8 Synthesis and final research positioning

The review indicates that the building blocks required for this project already exist separately: annotated contract corpora support clause classification, legal explainability research defines what a good legal explanation should contain, and Sri Lankan regulatory corpora provide a realistic, dated source of legal change. What is missing, and what this project contributes, is the connection between these three elements into a single evaluated system: a clause classifier whose explanations are traceable to specific regulatory provisions, and a monitoring mechanism that recognises when those provisions change. This distinguishes the proposed work from applied contract risk tools, which appear to stop at point-in-time risk scoring (Gangadhar, 2026; Prakash, Ramesh and Nair, n.d.), and from legal explainability research, which has not yet been tested against a live regulatory knowledge base (Mansi, Karusala and Riedl, 2025; Prajescu and Confalonieri, 2025).

2.9 Proposed conceptual architecture

Figure 2.2 shows the high-level architecture. Contracts and regulatory documents enter through separate ingestion paths, are segmented into clauses and provisions respectively, and are linked through a retrieval layer. A shared classification and explanation layer produces a risk flag, a confidence score and a provision-linked explanation for each clause, while a temporal monitor continuously compares the linked provision’s current version against the version that was active at the time of the original review, raising a compliance-drift alert when a difference is detected.

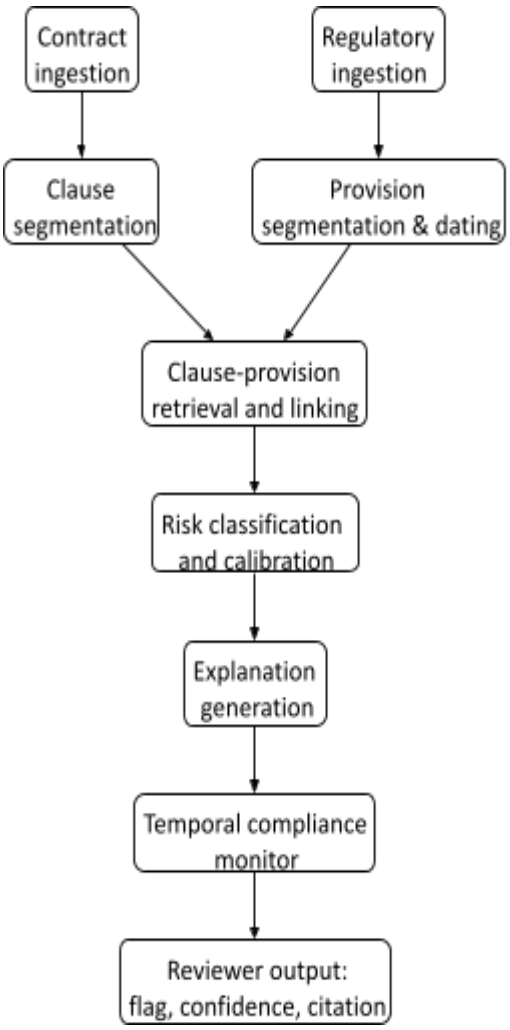


Figure 2.2: Proposed conceptual architecture for the explainable, temporally-aware contract intelligence system

2.10 Chapter summary

This chapter established that reliable, legally useful contract intelligence must combine clause classification, provision-traceable explanation and temporal regulatory monitoring, and that these three elements have not previously been evaluated together. The review justified the choice of CUAD, LEDGAR and ContractNLI as contract-level data sources, the Sri Lankan Acts, Bills and Extraordinary Gazette collections as the regulatory knowledge base, and an argumentation-informed explanation format supported by the legal explainability literature. Chapter 3 now translates these findings into the detailed research design, technical pipeline, requirements, architecture and project plan.

Chapter 3 - Methodology

3.1 Chapter overview

This chapter defines how the research questions from Chapter 1 will be answered. It begins with the research methodology expressed through Saunders' Research Onion, then presents the research design and hypotheses, the end-to-end technical pipeline, the requirements and architecture, the data management plan, the development approach, and the evaluation methodology.
It closes with the project schedule, resources, risk analysis and preliminary feasibility evidence.

3.2 Research methodology using Saunders' Research Onion

A pragmatic philosophy is appropriate because the project evaluates whether a technical artefact, the contract intelligence system, produces reliable and legally useful evidence under clearly defined conditions, rather than testing a single universal theory of legal reasoning. The approach is largely deductive, since hypotheses derived from the literature review are tested through controlled experiments, although the explanation format will be refined iteratively as feedback becomes available. The methodological choice is a quantitative, mono-method design centred on model benchmarking, supplemented by a small qualitative check with an available legal reviewer if ethical approval and access permit. The strategy is design-and-evaluation research, and the time horizon is cross-sectional, since all experiments will be conducted on existing, retrospective contract and regulatory data during the project timeline (Saunders, Lewis and Thornhill, 2019).

3.3 Research design and hypotheses

For RQ1, the independent variable is the clause representation strategy, comparing a TF-IDF classical baseline, a general-purpose transformer encoder, and a legal-domain fine-tuned transformer. The dependent variables are macro-F1 and class specific sensitivity across the CUAD risk categories. H1 states that the legal-domain fine-tuned transformer will outperform both baselines under the same data split. For RQ2, explanation methods are compared on faithfulness, measured through a deletion and insertion protocol, and on stability, measured through explanation change under paraphrased clause text, with H2 stating that the argumentationinformed explanation will be judged at least as faithful as attribution-based explanation while providing additional legal traceability. For RQ3, the temporal monitor is tested using simulated regulatory amendments drawn from the dated

version history in the Acts and Extraordinary Gazette collections, with H3 stating that the monitor will detect a majority of simulated compliance-affecting amendments at an acceptable false alarm rate. Ablation studies will reTable 3.1: Research Onion layers applied to this project

Layer	Selected choice	Project-specific justification
Philosophy	Pragmatism	Focuses on building and evaluating an artefact against practical and legal usefulness criteria
Approach	Deductive with iterative refinement	Tests literature-derived hypotheses while refining the explanation format
Methodological choice	Quantitative, optional qualitative supplement	Experiments provide the primary evidence; reviewer feedback is secondary
Strategy	Design science / experimental evaluation	Builds an artefact and evaluates its predictive, explanatory and temporal performance
Time horizon	Crosssectional	Experiments run on existing retrospective data during project execution
Techniques	Classification training, calibration, faithfulness testing, drift simulation	Directly answer RQ1 through RQ4

move the temporal monitor, the calibration step, and the argumentation layer in turn to isolate their individual contribution.

3.4 Proposed technical methodology and implementation pipeline

3.4.1 Data ingestion and governance

CUAD, LEDGAR and ContractNLI will be ingested through their official Hugging Face or GitHub releases, and the Sri Lanka Acts, Bills and Extraordinary Gazette collections will be ingested through their MIT-licensed GitHub data branches (Senaratna et al., 2026). A dataset manifest will record source, licence, version and split assignment for every document, and raw data will be kept read-only, with all derived artefacts versioned separately.

3.4.2 Data quality assessment

Each contract and regulatory document will be checked for extraction completeness, encoding errors and duplicate content, following the same duplicate-detection logic already applied in the source corpora. Documents that fail minimum quality checks, such as unreadable text extraction, will be logged and excluded with a documented reason rather than silently dropped.

3.4.3 Pre-processing

Contracts will be segmented into clauses aligned with the existing CUAD and LEDGAR annotation boundaries, and regulatory documents will be segmented into dated sections. All text will be lower-cased only where required by a specific model, since case can carry legal meaning in defined terms, and all transforms fitted on the training partition will be reused, unchanged, on the validation and test partitions to avoid leakage.

3.4.4 Baseline construction

Baselines will include a TF-IDF with logistic regression classifier, a general-purpose pretrained transformer without legal fine-tuning, and a snapshot-only compliance checker that ignores provision versioning entirely, providing a direct comparison point for the value added by temporal monitoring.

3.4.5 Proposed model

The proposed model uses a legal-domain fine-tuned transformer encoder to produce a clause representation, a classification head trained with class-weighted loss for the risk categories, and a separate retrieval component that ranks candidate regulatory provisions for each clause using dense embedding similarity combined with legal subject tag filtering.

3.4.6 Risk calibration and escalation mechanism

Raw classifier outputs will be calibrated using temperature scaling on the validation partition, and a defer threshold, also selected on validation data only, will determine whether a clause is flagged automatically, escalated for mandatory review, or passed without comment, following standard selective-prediction practice.

3.4.7 Explanation generation

For each flagged clause, the system will generate an attribution-based explanation highlighting influential text, together with an argumentation-informed explanation that

states the risk category, the linked regulatory provision, its effective date, and a short justification connecting the clause wording to that provision.

3.4.8 Reproducibility and experiment tracking

Every experiment run will store the dataset version, split, random seed, hyperparameters, model checkpoint, evaluation metrics and system environment details, enabling any reported result to be reproduced from its saved configuration.

3.5 Requirements engineering

3.5.1 Stakeholder analysis

Primary stakeholders are legal and compliance reviewers who would use the system's output. Secondary stakeholders include data custodians responsible for the public legal document collections, system administrators, and the wider regulatory environment represented by data protection and electronic transaction law. Researchers and future maintainers form an additional stakeholder group responsible for extending the system beyond the current project scope.

3.5.2 Requirement elicitation and evidence

Requirements are derived primarily from the literature review, the documented structure of the CUAD, LEDGAR and Sri Lankan regulatory datasets, and the workflow description in Section 1.2.3. If access to a legal reviewer becomes available and ethical approval is granted, a short structured review of the explanation format will be used to validate whether the output is understandable and legally useful; this will only take place after approval and is not assumed as a certainty in this proposal.

3.5.3 Functional and non-functional requirements

Table 3.2: Selected functional and non-functional requirements

ID	Priority	Requirement	Verification
FR-M1	Must	Accept a contract document and segment it into individual clauses	Pipeline test with valid and malformed inputs
FR-M2	Must	Return a predicted risk category, a calibrated confidence score and an escalation decision for each clause	Automated output-schema test

FR-M3	Must	Return an explanation for each flagged clause, including the linked regulatory provision and its effective date	Explanation module test
FR-S1	Should	Re-evaluate previously reviewed contracts when a linked regulatory provision changes	Simulated amendment regression test
NFR-R1	Must	Reproduce an experiment from its saved configuration and seed	Repeat-run comparison
NFR-P1	Must	Avoid storing personal or confidential contract content beyond the active processing session	Data-flow and access-control review

3.6 System and software design

The system separates into five components: a document manager handling ingestion and segmentation of both contracts and regulatory text, a classification service, a retrieval and linking service connecting clauses to provisions, an explanation service, and an experiment and evaluation manager. A reviewer submits a contract, receives clause-level flags with linked citations, and can request a re-check of a previously submitted contract, which triggers the temporal compliance monitor rather than a full re-classification. This separation allows each research component to be replaced independently during ablation studies, consistent with the architecture in Figure 2.2.

3.7 Data requirements and management plan

CUAD will serve as the primary source of clause-level training and evaluation labels, LEDGAR will provide additional scale for provision classification, and ContractNLI will support compliancestyle entailment checks. The Sri Lanka Acts collection will be the primary regulatory knowledge base, with the Extraordinary Gazette collections used to reconstruct dated amendments for the temporal compliance experiments. A fixed, document-level split will be created before model development so that no document appears in more than one partition. Derived files will carry provenance metadata linking each output back to its source document and processing configuration, and access to any locally added example contracts will be restricted to processing needed for the project.

3.8 Development methodology

An iterative Agile approach, organised into two-week sprints, is appropriate because model, retrieval and explanation design decisions are expected to evolve through experimentation. Each sprint will produce a testable increment, such as the clause segmentation pipeline, the baseline classifier, the retrieval and linking layer, or the temporal monitor, with a clear definition of done and version-controlled code reviewed before merging.

3.9 Evaluation methodology

3.9.1 Dataset splits and validation protocol

All splits will be created at the document level so that no contract or regulatory document contributes to more than one partition. Hyperparameters, calibration temperature and the escalation threshold will be selected on the validation partition only, and the test partition will be evaluated once after the design is fixed.

3.9.2 Baselines and ablation studies

The full system will be compared against the TF-IDF baseline, the general-purpose transformer without legal fine-tuning, and the snapshot-only compliance checker described in Section 3.4.4. Ablations will remove the calibration step, the argumentation-informed explanation, and the temporal monitor in turn to isolate each component's contribution.

3.9.3 Metrics and analysis

Classification performance will be reported using macro-F1 and class-specific precision and recall. Calibration will be reported using expected calibration error and a reliability diagram. Explanation quality will be reported using a deletion and insertion faithfulness score and a paraphrase-based stability score. Temporal compliance detection will be reported as precision and recall over simulated regulatory amendments, together with a small set of manually reviewed example cases.

3.9.4 Research question to evaluation traceability

RQ	Experiment	Evidence
RQ1	Legal-domain transformer vs. classical and general-purpose baselines	Macro-F1, classspecific precision/recall

RQ2	Attribution-based vs. argumentation-informed explanation	Faithfulness score, stability score
RQ3	Simulated regulatory amendment injection	Drift-detection precision and recall
RQ4	Structured reviewer feedback, subject to ethical approval	Qualitative usefulness rating

Table 3.3: Research question to evaluation traceability

3.10 Project management methodology

The project will use a Kanban board with work-in-progress limits and fortnightly milestone reviews. Work packages are organised as literature and data readiness, baseline development, the proposed model and retrieval layer, explanation and calibration development, temporal monitor development, and evaluation and writing, with each task linked to a specific objective.

3.11 Resources and new technical skills

The project requires a GPU-enabled machine sufficient for fine-tuning a mid-sized transformer encoder, standard Python libraries for natural language processing and machine learning, version control, and an experiment tracking tool. New skills to be developed include legal-domain

Period	Activities	Milestone evidence
Weeks 1–3	Confirm data access, finalise requirements, refine literature matrices	Approved scope and dataset decision
Weeks 4–6	Build data pipeline and baseline classifiers	Validated pipeline, baseline results
Weeks 7–10	Implement legal-domain classifier and clause provision retrieval	Model checkpoints, retrieval evaluation
Weeks 11–13	Implement calibration and explanation generation	Faithfulness and stability results
Weeks 14–16	Implement and evaluate the temporal compliance monitor	Drift-detection evaluation
Weeks 17–18	Build demonstrator interface, run ablation studies	Final experiment package
Weeks 19–20	Analysis, writing, verification and submission preparation	Complete dissertation and artefacts

Table 3.4: Indicative schedule and milestones

transformer fine-tuning, calibration and selective-prediction techniques, and argumentationstyle explanation generation. A fallback plan using a smaller, more efficient encoder is available if fine-tuning the larger model proves infeasible within the available compute budget.

3.12 Risk analysis and mitigation

Risk	Likelihood	Impact	Prevention and contingency
Clause provision linking is inaccurate	Medium	High	Manually verify a sample of links early; fall back to a smaller, curated subset of provisions if needed
Legal-domain transformer training exceeds compute budget	Medium	Medium	Use a smaller encoder as a fallback; profile memory use early
No access to a legal reviewer for RQ4 feedback	Medium	Low	Treat RQ4 as optional and clearly scope the claim if feedback is unavailable
Document-level data leakage across splits	Low	Critical	Automated split-integrity checks before any training run
Generative AI misuse during writing or coding	Low	High	Maintain an AI-use log and verify all AI-assisted text and code before use

Table 3.5: Key project risks and mitigation

3.13 Preliminary proof of concept and feasibility evidence

An early feasibility check has confirmed that the CUAD annotation files and a small sample of the Sri Lanka Acts collection can both be loaded and parsed successfully, and that a simple TF-IDF baseline can be trained on a small CUAD subset without errors. This demonstrates technical feasibility only and is not used to claim any comparative result. The next feasibility step will confirm that a single clause can be linked to a plausible regulatory provision end to end, before the full pipeline is built.

3.14 Chapter summary

This chapter defined a pragmatic, design-and-evaluation methodology supported by documentlevel experiments, defined baselines, targeted ablations and a dedicated temporal compliance evaluation protocol. It specified the end-to-end pipeline, requirements, architecture, schedule, resources and risk controls needed to deliver the project. Chapter 4 now addresses the legal, ethical, social and professional conditions that govern this project, including the responsible use of Generative AI tools.

Chapter 4 - LESP

4.1 Chapter overview

This chapter evaluates the social, legal, ethical and professional implications of building an explainable contract intelligence system that processes potentially sensitive commercial documents and references binding legal instruments. It addresses fairness and accessibility, data protection and licensing, transparency and human oversight, and professional responsibility in reporting, and it closes with a dedicated account of how Generative AI tools will be used responsibly throughout the project.

4.2 Social issues

A model trained mainly on international, English-language contract corpora such as CUAD and LEDGAR may perform less reliably on contracts drafted in Sri Lankan legal style or on bilingual documents, and this limitation will be reported explicitly rather than implied away. The system is intended to support, not replace, a human reviewer, and the demonstrator interface will clearly present each output as a flag for review rather than a final legal determination, reducing the risk of automation bias among less experienced users.

4.3 Legal issues

All datasets used in this project carry permissive public licences, CC BY 4.0 for CUAD and LEDGAR, and MIT for the Sri Lankan Acts, Bills and Extraordinary Gazette collections (Hendrycks et al., 2021; Tuggener et al., 2020; Senaratna et al., 2026), and their terms will be followed and acknowledged in the data management plan. Any locally sourced example contracts will be anonymised or based on non-confidential templates before use. Processing of contract data will be considered under Sri Lanka's Personal Data Protection Act, No. 9 of 2022, and the legal recognition of any electronically signed source material will be considered under the Electronic Transactions Act, No. 19 of 2006. Confidential or restricted material will not be submitted to any third-party Generative AI service.

4.4 Ethical issues

Explanations produced by the system will be evaluated for faithfulness rather than presented as self-evidently correct, and the system will communicate confidence explicitly rather than forcing an uncertain clause into a falsely confident output. All reported experimental results

will come from preserved logs and configurations rather than being reconstructed or estimated after the fact, and no result will be generated or altered by a Generative AI tool.

4.5 Professional issues

The system is framed throughout as a decision-support prototype rather than a source of binding legal advice, and this distinction will be stated clearly in the demonstrator and the final dissertation. Negative or inconclusive results, including cases where the temporal compliance monitor fails to detect a simulated amendment, will be reported honestly. Any code adapted from external sources or AI-assisted suggestions will be reviewed, tested and attributed appropriately.

4.6 Responsible use of Generative AI tools

4.6.1 Intended and permitted use

Generative AI tools may be used to brainstorm literature search terms, to check the clarity of sentences that have already been drafted by the candidate, and to suggest test-case ideas that are then independently reviewed and extended before being added to the test suite.

4.6.2 Prohibited or high-risk use

Generative AI tools will not be used to fabricate citations, invent experimental results, generate synthetic legal opinions presented as authoritative, or produce code or text submitted as original work without review. Confidential contract content, if any is obtained, will not be entered into any public Generative AI service.

4.6.3 Verification and human accountability

Every citation used in this proposal has been checked against its source file or official page, every suggested code fragment will be run and tested before being kept, and the candidate remains fully accountable for every design and implementation decision in the project.

4.6.4 Privacy, confidentiality and intellectual property

Only synthetic or clearly non-confidential example text will be used when consulting a Generative AI tool for debugging assistance, and the licence terms of the CUAD, LEDGAR, ContractNLI and Sri Lankan regulatory datasets will be respected throughout.

4.6.5 Transparency, disclosure and AI-use register

A brief AI-use register will be maintained recording the tool used, the date, the purpose, and how the output was verified or modified, summarised in the declaration at the front of this document and retained in full for supervisor review on request.

4.6.6 Example disclosure statement

"Generative AI tools were used in a limited supporting capacity to brainstorm search terms and review the clarity of student-written text. All outputs were critically reviewed, every citation was verified against its original source, and no AI-generated results, fabricated references or confidential data were used. The student remains fully responsible for the content, design and conclusions of this proposal."

4.6.7 Project-specific Generative AI risk assessment

Risk	Possible consequence	Control
Hallucinated citations or legal claims	Invalid literature base or misleading legal statements	Verify every citation and legal claim against the original source before use
Insecure or incorrect AI-suggested code	Vulnerable or invalid experiments	Line-by-line review, testing and version control before merging
Confidentiality breach	Unauthorised disclosure of contract or institutional data	Use no confidential input; synthetic examples only

Table 4.1: Generative AI risk assessment for this project

4.7 Chapter summary

This chapter has shown that the project's social, legal, ethical and professional risks are managed through explicit limitation reporting, respect for dataset licences and Sri Lankan

data protection law, faithful and honestly reported explanations, and a disclosed, verified and limited use of Generative AI tools throughout the research process.

References

- Anon. (2026) ‘Debate: Evidence-based AI risk assessment for public policy’, [PDF], accessed from project research folder, RESEARCH PAPERS/Final Research Papers/files.
- Anon. (n.d.) ‘Leveraging Generative AI and Natural Language Processing for Legal Risk Management: Transforming Compliance, Contract Analysis and Decision-Making in the Legal Domain’, [PDF], accessed from project research folder, RESEARCH PAPERS/Final Research Papers/New.
- Electronic Transactions Act, No. 19 of 2006 (Sri Lanka).
- European Union (2016) *Regulation (EU) 2016/679 (General Data Protection Regulation)*. Official Journal of the European Union.
- European Union (2024) *Regulation (EU) 2024/1689 (Artificial Intelligence Act)*. Official Journal of the European Union.
- Gangadhar, S. (2026) ‘AI-Powered Legal Contract Risk Analyzer’, [PDF], accessed from project research folder, RESEARCH PAPERS/Final Research Papers/files.
- Hacker, P., Krestel, R., Grundmann, S. and Naumann, F. (2020) ‘Explainable AI under contract and tort law: legal incentives and technical challenges’, *Artificial Intelligence and Law*, 28(4), pp. 415–439.
- Hendrycks, D., Burns, C., Chen, A. and Ball, S. (2021) ‘CUAD: An Expert-Annotated NLP Dataset for Legal Contract Review’, in *Proceedings of the Neural Information Processing Systems (NeurIPS) Track on Datasets and Benchmarks*.
- Keerthana (n.d.) conference paper on AI-based legal and contract analysis, KSIT, [PDF], accessed from project research folder, RESEARCH PAPERS/Final Research Papers/files.
- Koreeda, Y. and Manning, C.D. (2021) ‘ContractNLI: A Dataset for Document-level Natural Language Inference for Contracts’, in *Findings of the Association for Computational Linguistics: EMNLP 2021*.
- Kostopoulos, G., et al. (2024) ‘Explainable Artificial Intelligence-Based Decision Support Systems: A Recent Review’, [PDF], accessed from project research folder, RESEARCH PAPERS/Final Research Papers/files.

- Longo, L., et al. (2024) ‘Explainable Artificial Intelligence (XAI) 2.0: A Manifesto of Open Challenges and Interdisciplinary Directions’, [PDF], accessed from project research folder, RESEARCH PAPERS/Final Research Papers/files.

- Mafleh, R., et al. (2026) ‘Exploring the Legal Impact of Artificial Intelligence on Contract Interpretation and Execution’, [PDF], accessed from project research folder, RESEARCH PAPERS/Final Research Papers/files.
- Mansi, G., Karusala, N. and Riedl, M. (2025) ‘Legally-Informed Explainable AI’, arXiv preprint arXiv:2504.10708.
- Personal Data Protection Act, No. 9 of 2022 (Sri Lanka).
- Prajescu, A.I. and Confalonieri, R. (2025) ‘Argumentation-Based Explainability for Legal AI: Comparative and Regulatory Perspectives’, arXiv preprint arXiv:2510.11079.
- Prakash, V., Ramesh, S. and Nair, A. (n.d.) ‘AI-Powered Regulatory Compliance Checker for Contracts’, [PDF], accessed from project research folder, RESEARCH PAPERS/Final Research Papers/files.
- Saeed, W. and Omlin, C. (2021) ‘Explainable AI (XAI): A Systematic Meta-Survey of Current Challenges and Future Opportunities’, arXiv preprint arXiv:2111.06420.
- Saunders, M., Lewis, P. and Thornhill, A. (2019) *Research Methods for Business Students*, 8th edn. Harlow: Pearson.
- Senaratna, N., et al. (2026) ‘Sri Lanka Document Datasets: A Large-Scale, Multilingual Resource for Law, News, and Policy’, accessed via project research summary and official GitHub/Hugging Face repositories (nuuuwan/lk acts, nuuuwan/lk bills, nuuuwan/lk legal docs, nuuuwan/lk appeal court judgements, nuuuwan/lk supreme court ¨judgements).
- Tuggener, D., von Daniken, P., Peetz, T. and Cieliebak, M. (2020) ‘LEDGAR: A Large-¨ Scale Multi-label Corpus for Text Classification of Legal Provisions in Contracts’, in *Proceedings of the 12th Language Resources and Evaluation Conference (LREC 2020)*.

Appendices

Appendix	Recommended content
A	Detailed literature search strategy and screening log
B	Full-size concept map
C	Literature survey matrix (all reviewed papers)
D	Research gap analysis matrix
E	Dataset comparison matrix and data dictionary
F	Stakeholder analysis and elicitation notes
G	Full functional and non-functional requirements
H	Use-case descriptions and sequence diagrams
I	Detailed architecture and data-flow diagrams
J	Detailed Gantt chart and backlog
K	Full risk register and issue log
L	Generative AI use declaration and AI-use register
M	Preliminary proof-of-concept logs and sample outputs
N	Ethics clearance material, where applicable

Table 2: Appendix contents overview